

Thiagarajar College of Engineering

Name: gokul g
Email: gokulg@student.tce.edu
Roll no: 2403917710421020
Phone: 6369120591
Branch: TCE
Department: CSE - A
Batch: 2028
Degree: BE CSE

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2024_28_III_22CS370_Data Structures Lab_A

TCE_22CS370_Week 4_COD

Attempt : 1
Total Mark : 30
Marks Obtained : 30

Section 1 : Coding

1. Problem Statement

Sudha is taking a course on polynomials, and her instructor has assigned her the task of writing a program to perform addition on two polynomials represented as linked lists.

Sudha needs your help to implement the program.

Input Format

The first line of input consists of an integer n , representing the number of terms in the first polynomial.

The following n lines of input consist of two integers each: the coefficient and the exponent of the term in the first polynomial.

The next line of input consists of an integer m , representing the number of terms

in the second polynomial.

The following m lines of input consist of two integers each: the coefficient and the exponent of the term in the second polynomial.

Output Format

The output prints the resulting polynomial after adding the given polynomials.

Refer to the sample output for the exact format.

Sample Test Case

Input: 3

3 2

4 1

5 0

3

2 3

1 2

2 0

Output: $(2x^3) + (4x^2) + (4x^1) + (7x^0)$

Answer

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
struct node{
```

```
    int coeff,exp;
```

```
    struct node*next;
```

```
};
```

```
struct node* insert(struct node* head,int c,int e){
```

```
    struct node* newnode=(struct node*)malloc(sizeof(struct node));
```

```
    newnode->coeff=c;
```

```
    newnode->exp=e;
```

```
    newnode->next=NULL;
```

```
    if(!head) return newnode;
```

```
    struct node* cur=head;
```

```
    while(cur->next) cur=cur->next;
```

```
    cur->next=newnode;
```

```
    return head;
```

```
}
```

```

struct node* add(struct node* p1, struct node* p2){
    struct node* res=NULL;
    while(p1!=NULL && p2!=NULL){
        if(p1->exp > p2->exp){
            res=insert(res,p1->coeff,p1->exp);
            p1=p1->next;
        }
        else if(p1->exp < p2->exp){
            res=insert(res,p2->coeff,p2->exp);
            p2=p2->next;
        }
        else{
            int sum=p1->coeff+p2->coeff;

            res=insert(res,sum,p1->exp);
            //res=insert(res,p1->coeff+p2->coeff,p1->exp);
            p1=p1->next;
            p2=p2->next;
        }
    }
    while(p1!=NULL){
        res=insert(res,p1->coeff,p1->exp);
        p1=p1->next;
    }
    while(p2!=NULL){
        res=insert(res,p2->coeff,p2->exp);
        p2=p2->next;
    }
    return res;
}

```

```

void print(struct node* p){
    int fi=1;
    while(p){
        if(!fi) printf(" + ");
        printf("(%dx^%d)",p->coeff,p->exp);
        fi=0;p=p->next;
    }
    printf("\n");
}

```

```

int main(){

```

```

int n,m,c,e;
struct node* p1=NULL,*p2=NULL;
scanf("%d",&n);
for(int i=0;i<n;i++){
    scanf("%d %d",&c,&e);
    p1=insert(p1,c,e);
}
scanf("%d",&m);
for(int i=0;i<m;i++){
    scanf("%d %d",&c,&e);
    p2=insert(p2,c,e);
}
print(add(p1,p2));
return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Timothy wants to evaluate polynomial expressions for his mathematics homework. He needs a program that allows him to input the coefficients of a polynomial based on its degree and compute the polynomial's value for a given input of x . Implement a function that takes the degree, coefficients, and the value of x , and returns the evaluated result of the polynomial.

Example

Input:

degree of the polynomial = 2

coefficient of x^2 = 13

coefficient of x^1 = 12

coefficient of x^0 = 11

$x = 1$

Output:

36

Explanation:

Calculate the value of $13x^2$: $13 * 12 = 13$.

Calculate the value of $12x^1$: $12 * 11 = 12$.

Calculate the value of $11x^0$: $11 * 10 = 11$.

Add the values of x^2 , x^1 , and x^0 together: $13 + 12 + 11 = 36$.

Input Format

The first line of input consists of an integer representing the degree of the polynomial.

The second line consists of an integer representing the coefficient of x^2 .

The third line consists of an integer representing the coefficient of x^1 .

The fourth line consists of an integer representing the coefficient of x^0 .

The fifth line consists of an integer representing the value of x , at which the polynomial should be evaluated.

Output Format

The output is an integer value obtained by evaluating the polynomial at the given value of x .

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

13

12

11

1

Output: 36

Answer

```

#include<stdio.h>
#include<stdlib.h>
#include<math.h>
struct node{
    int co;
    int p;
    struct node* next;
};
struct node* create(int co,int p){
    struct node* newnode=(struct node*)malloc(sizeof(struct node));
    newnode->co=co;
    newnode->p=p;
    newnode->next=NULL;
    return newnode;
}void insertatend(struct node** head,int co,int p){
    struct node* newnode=create(co,p);
    if(*head==NULL) *head=newnode;
    else {
        struct node*temp=*head;
        while(temp->next!=NULL){
            temp=temp->next;
        }
        temp->next=newnode;
    }
}
int evaluate(struct node* head,int x){
    int r=0;
    while(head!=NULL){
        r+=head->co*pow(x,head->p);
        head=head->next;
    }
    return r;
}
int main(){
    int d;
    scanf("%d",&d);
    struct node* head=NULL;
    for(int i=d;i>=0;i--){
        int co;
        scanf("%d",&co);
        insertatend(&head,co,i);
    }
}

```

```
}  
int x;  
scanf("%d",&x);  
int a=evaluate(head,x);  
printf("%d",a);  
return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

John is working on a math processing application, and his task is to simplify polynomials entered by users. The polynomial is represented as a linked list, where each node contains two properties:

Coefficient of the term.

Exponent of the term.

John's goal is to combine all the terms that have the same exponent, effectively simplifying the polynomial.

Input Format

The first line of input consists of an integer representing the number of terms in the polynomial.

The next n lines of input consist of two integers, representing the coefficient and exponent of the polynomial in each line separated by space.

Output Format

The first line of output prints the original polynomial in the format ' $cx^e + cx^e + \dots$ ' (where c is the coefficient and e is the exponent of each term).

The second line of output displays the simplified polynomial in the same format as the original polynomial.

If the polynomial is 0, then only '0' will be printed.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

5 2

3 1

6 2

Output: Original polynomial: $5x^2 + 3x^1 + 6x^2$

Simplified polynomial: $11x^2 + 3x^1$

Answer

```
#include<stdio.h>
#include<stdlib.h>
struct node{
    int c;
    int p;
    struct node* n;
};
void ins(struct node** s,int c,int p){
    struct node* t=*s;
    while(t!=NULL){
        if(t->p==p){
            t->c+=c;
            return;
        }
        t=t->n;
    }
    struct node* nn=(struct node*)malloc(sizeof(struct node));
    nn->c=c;
    nn->p=p;
    nn->n=NULL;
    if(*s==NULL){
        *s=nn;
        return;
    }
    t=*s;
    while(t->n!=NULL){
        t=t->n;
    }
    t->n=nn;
}
```



```

void dis(struct node* s){
    struct node* t=s;
    while(t!=NULL){
        printf("%dx^%d",t->c,t->p);
        if(t->n!=NULL){
            printf(" + ");
        }
        t=t->n;
    }
}

int main(){
    int n,i,p,c;
    struct node* s=NULL;
    scanf("%d",&n);
    printf("Original polynomial: ");
    for(int i=0;i<n;i++){
        scanf("%d %d",&c,&p);
        printf("%dx^%d",c,p);
        if(i<n-1){
            printf(" + ");
        }
        ins(&s,c,p);
    }
    printf("\n");
    printf("Simplified polynomial: ");
    dis(s);
    return 0;
}

```

Status : Correct

Marks : 10/10